

Assignment No. 11 — Indexing and Query Optimization

Q1) Create a MySQL database named SalesDB and a table named OrderData.

OrderData Table

Column Name	Data Type	Constraint
OrderID	INT	PRIMARY KEY
CustomerName	VARCHAR(50)	NOT NULL
Email	VARCHAR(80)	NOT NULL
City	VARCHAR(30)	NOT NULL
ProductCategory	VARCHAR(30)	NOT NULL
OrderDate	DATE	NOT NULL
OrderAmount	DECIMAL(10,2)	NOT NULL
OrderStatus	VARCHAR(20)	NOT NULL

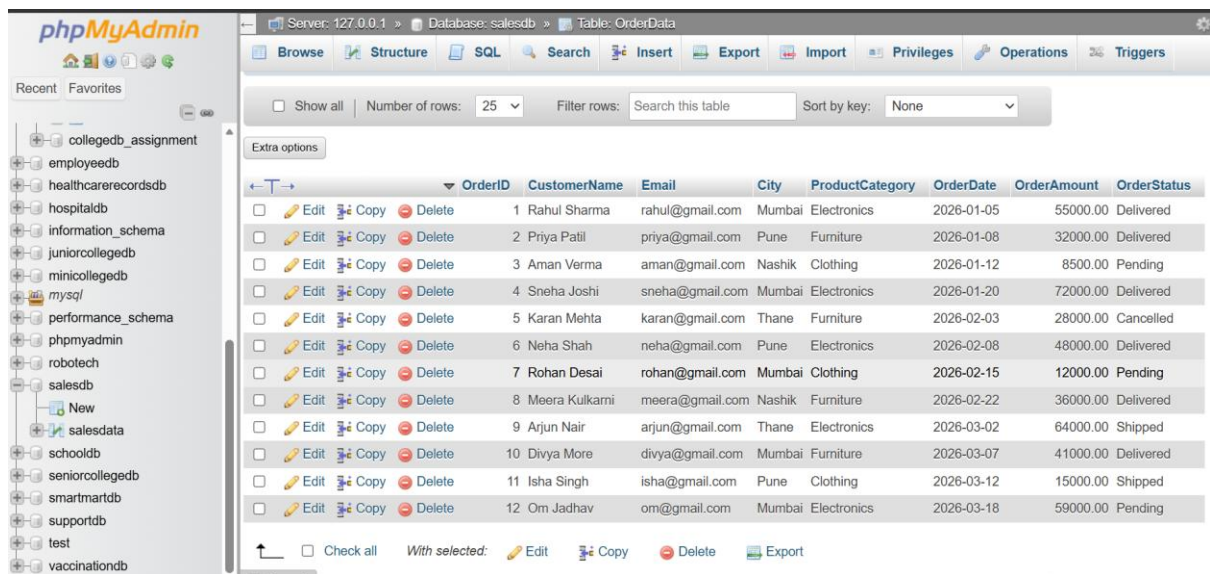
Insert at least 12 records into the table.

```
CREATE DATABASE SalesDB;
```

```
USE SalesDB;
```

```
CREATE TABLE OrderData (  
    OrderID INT PRIMARY KEY,  
    CustomerName VARCHAR(50) NOT NULL,  
    Email VARCHAR(80) NOT NULL,  
    City VARCHAR(30) NOT NULL,  
    ProductCategory VARCHAR(30) NOT NULL,  
    OrderDate DATE NOT NULL,  
    OrderAmount DECIMAL(10,2) NOT NULL,  
    OrderStatus VARCHAR(20) NOT NULL  
);
```

```
INSERT INTO OrderData
(OrderID, CustomerName, Email, City, ProductCategory,
OrderDate, OrderAmount, OrderStatus)
VALUES
(1, 'Rahul Sharma', 'rahul@gmail.com', 'Mumbai',
'Electronics', '2026-01-05', 55000.00, 'Delivered'),
(2, 'Priya Patil', 'priya@gmail.com', 'Pune',
'Furniture', '2026-01-08', 32000.00, 'Delivered'),
(3, 'Aman Verma', 'aman@gmail.com', 'Nashik',
'Clothing', '2026-01-12', 8500.00, 'Pending'),
(4, 'Sneha Joshi', 'sneha@gmail.com', 'Mumbai',
'Electronics', '2026-01-20', 72000.00, 'Delivered'),
(5, 'Karan Mehta', 'karan@gmail.com', 'Thane',
'Furniture', '2026-02-03', 28000.00, 'Cancelled'),
(6, 'Neha Shah', 'neha@gmail.com', 'Pune',
'Electronics', '2026-02-08', 48000.00, 'Delivered'),
(7, 'Rohan Desai', 'rohan@gmail.com', 'Mumbai',
'Clothing', '2026-02-15', 12000.00, 'Pending'),
(8, 'Meera Kulkarni', 'meera@gmail.com', 'Nashik',
'Furniture', '2026-02-22', 36000.00, 'Delivered'),
(9, 'Arjun Nair', 'arjun@gmail.com', 'Thane',
'Electronics', '2026-03-02', 64000.00, 'Shipped'),
(10, 'Divya More', 'divya@gmail.com', 'Mumbai',
'Furniture', '2026-03-07', 41000.00, 'Delivered'),
(11, 'Isha Singh', 'isha@gmail.com', 'Pune',
'Clothing', '2026-03-12', 15000.00, 'Shipped'),
(12, 'Om Jadhav', 'om@gmail.com', 'Mumbai',
'Electronics', '2026-03-18', 59000.00, 'Pending');
```



Q2) Create the following MySQL indexes on the OrderData table:

Index Name	Column	Index Type
idx_city	City	Normal index
idx_email	Email	Unique index
idx_order_date	OrderDate	Normal index

CREATE INDEX idx_city

ON OrderData(City);

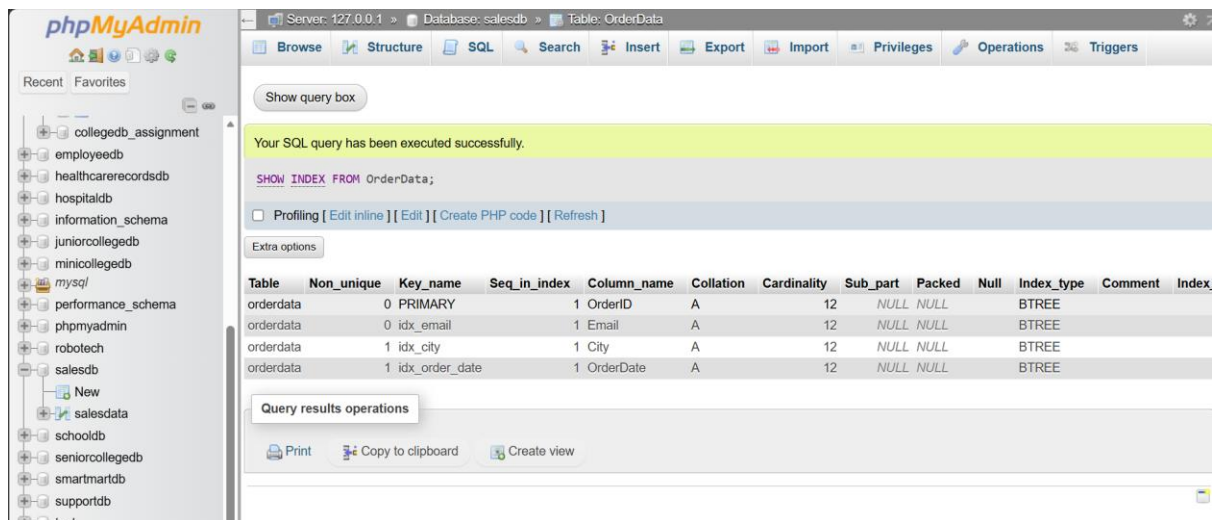
CREATE UNIQUE INDEX idx_email

ON OrderData(Email);

CREATE INDEX idx_order_date

ON OrderData(OrderDate);

SHOW INDEX FROM OrderData;



Q3) Create a composite index named idx_city_order_date.

The index should contain:

Index Position	Column
1	City
2	OrderDate

Write a query that searches for Mumbai orders within a specified date range.

```
CREATE INDEX idx_city_order_date
```

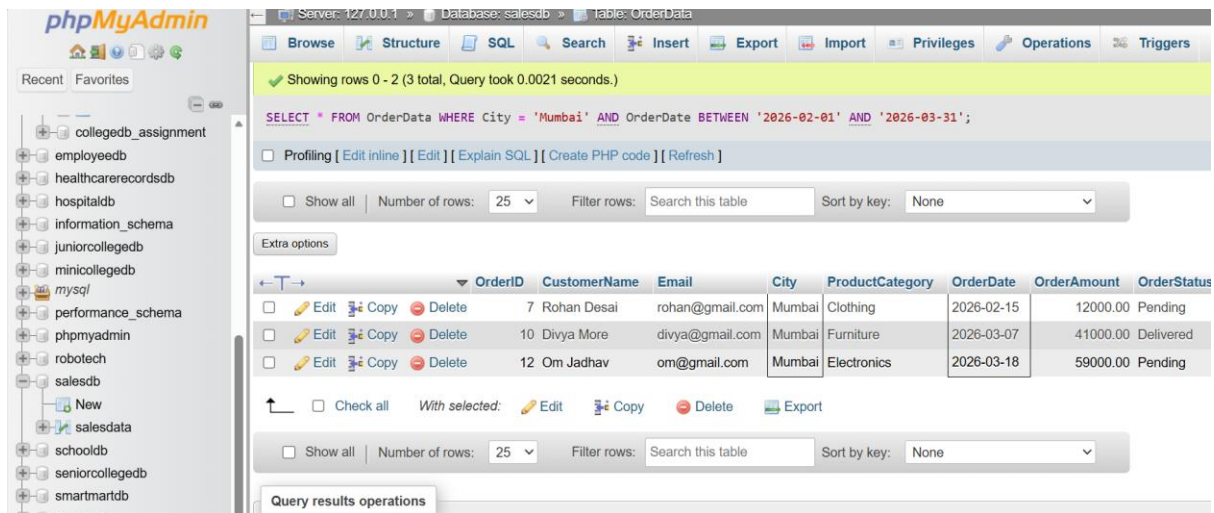
```
ON OrderData(City, OrderDate);
```

```
SELECT *
```

```
FROM OrderData
```

```
WHERE City = 'Mumbai'
```

```
AND OrderDate BETWEEN '2026-02-01' AND '2026-03-31';
```



Q4) Write a MySQL EXPLAIN query to analyze the following search:

SELECT *
FROM OrderData
WHERE City = 'Mumbai';

Record and explain the following output columns:

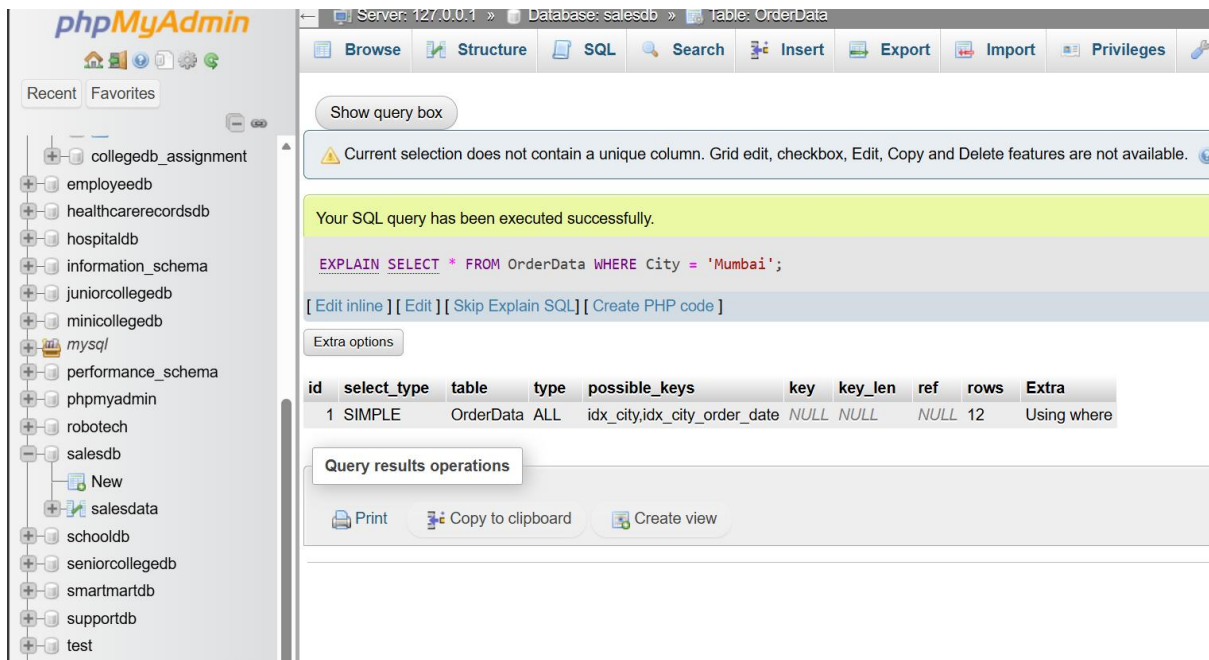
EXPLAIN Column
type
possible keys
key
rows
Extra

EXPLAIN

SELECT *

FROM OrderData

WHERE City = 'Mumbai';



Q5) Compare query performance before and after creating an index on City.

Perform the following steps:

Step	Task
1	Drop idx_city, if it exists
2	Run EXPLAIN on the City search
3	Create idx_city
4	Run the same EXPLAIN query again
5	Compare the selected index and estimated rows

DROP INDEX idx_city

ON OrderData;

EXPLAIN

SELECT *

FROM OrderData

WHERE City = 'Mumbai';

phpMyAdmin

Server: 127.0.0.1 » Database: salesdb » Table: OrderData

Browse Structure SQL Search Insert Export Import Privileges

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Show query box

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Your SQL query has been executed successfully.

[EXPLAIN](#) [SELECT](#) * FROM OrderData WHERE City = 'Mumbai';

[[Edit inline](#)] [[Edit](#)] [[Skip Explain SQL](#)] [[Create PHP code](#)]

Extra options

id	select_type	table	type	possible_keys	key	key_len	ref	rows	Extra
1	SIMPLE	OrderData	ALL	idx_city_order_date	NULL	NULL	NULL	12	Using where

Query results operations

[Print](#) [Copy to clipboard](#) [Create view](#)

CREATE INDEX idx_city

ON OrderData(City);

EXPLAIN

SELECT *

FROM OrderData

WHERE City = 'Mumbai';

phpMyAdmin

Server: 127.0.0.1 » Database: salesdb » Table: OrderData

Browse Structure SQL Search Insert Export Import Privileges

Show query box

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Your SQL query has been executed successfully.

EXPLAIN SELECT * FROM OrderData WHERE City = 'Mumbai';

[Edit inline] [Edit] [Skip Explain SQL] [Create PHP code]

Extra options

id	select_type	table	type	possible_keys	key	key_len	ref	rows	Extra
1	SIMPLE	OrderData	ALL	idx_city_order_date,idx_city	NULL	NULL	NULL	12	Using where

Query results operations

Print Copy to clipboard Create view

Q6) Rewrite the following query in a more index-friendly form: **SELECT * FROM OrderData WHERE YEAR(OrderDate) = 2026;**

SELECT *

FROM OrderData

WHERE YEAR(OrderDate) = 2026;

phpMyAdmin

Server: 127.0.0.1 » Database: salesdb » Table: OrderData

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

From: [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	OrderID	CustomerName	Email	City	ProductCategory	OrderDate	OrderAmount	OrderStatus
☐ Edit ☐ Copy ☐ Delete	1	Rahul Sharma	rahul@gmail.com	Mumbai	Electronics	2026-01-05	55000.00	Delivered
☐ Edit ☐ Copy ☐ Delete	2	Priya Patil	priya@gmail.com	Pune	Furniture	2026-01-08	32000.00	Delivered
☐ Edit ☐ Copy ☐ Delete	3	Aman Verma	aman@gmail.com	Nashik	Clothing	2026-01-12	8500.00	Pending
☐ Edit ☐ Copy ☐ Delete	4	Sneha Joshi	sneha@gmail.com	Mumbai	Electronics	2026-01-20	72000.00	Delivered
☐ Edit ☐ Copy ☐ Delete	5	Karan Mehta	karan@gmail.com	Thane	Furniture	2026-02-03	28000.00	Cancelled
☐ Edit ☐ Copy ☐ Delete	6	Neha Shah	neha@gmail.com	Pune	Electronics	2026-02-08	48000.00	Delivered
☐ Edit ☐ Copy ☐ Delete	7	Rohan Desai	rohan@gmail.com	Mumbai	Clothing	2026-02-15	12000.00	Pending
☐ Edit ☐ Copy ☐ Delete	8	Meera Kulkarni	meera@gmail.com	Nashik	Furniture	2026-02-22	36000.00	Delivered
☐ Edit ☐ Copy ☐ Delete	9	Arjun Nair	arjun@gmail.com	Thane	Electronics	2026-03-02	64000.00	Shipped
☐ Edit ☐ Copy ☐ Delete	10	Divya More	divya@gmail.com	Mumbai	Furniture	2026-03-07	41000.00	Delivered
☐ Edit ☐ Copy ☐ Delete	11	Isha Singh	isha@gmail.com	Pune	Clothing	2026-03-12	15000.00	Shipped
☐ Edit ☐ Copy ☐ Delete	12	Om Jadhav	om@gmail.com	Mumbai	Electronics	2026-03-18	59000.00	Pending

Check all With selected: Edit Copy Delete Export

SELECT

OrderID,

CustomerName,

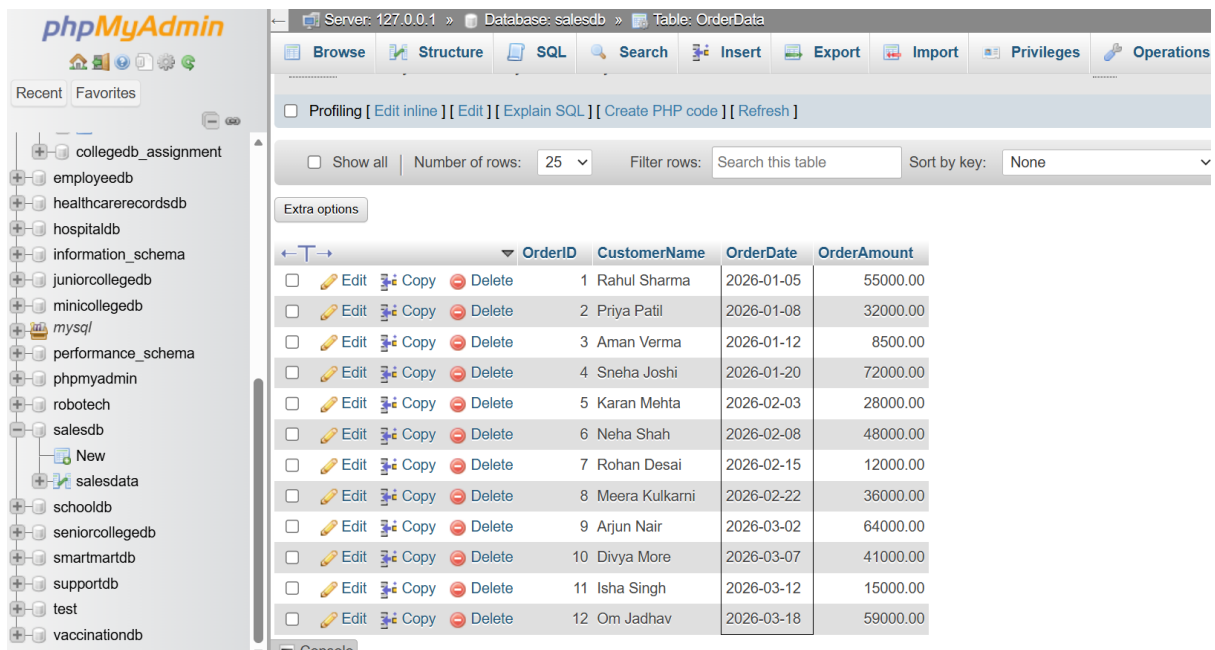
OrderDate,

OrderAmount

FROM OrderData

WHERE OrderDate >= '2026-01-01'

AND OrderDate < '2027-01-01';



The screenshot displays the phpMyAdmin interface. On the left, a sidebar shows a list of databases, with 'salesdb' selected. The main panel shows the 'OrderData' table structure and data. The table has four columns: OrderID, CustomerName, OrderDate, and OrderAmount. The data is filtered to show rows where OrderDate is greater than or equal to '2026-01-01' and less than '2027-01-01'. The table contains 12 rows of data.

OrderID	CustomerName	OrderDate	OrderAmount
1	Rahul Sharma	2026-01-05	55000.00
2	Priya Patil	2026-01-08	32000.00
3	Aman Verma	2026-01-12	8500.00
4	Sneha Joshi	2026-01-20	72000.00
5	Karan Mehta	2026-02-03	28000.00
6	Neha Shah	2026-02-08	48000.00
7	Rohan Desai	2026-02-15	12000.00
8	Meera Kulkarni	2026-02-22	36000.00
9	Arjun Nair	2026-03-02	64000.00
10	Divya More	2026-03-07	41000.00
11	Isha Singh	2026-03-12	15000.00
12	Om Jadhav	2026-03-18	59000.00